

1.1 Coulomb's Law and Superposition

Date: _____

- Two point charges, $q_1 = +3.0 \mu\text{C}$ and $q_2 = -5.0 \mu\text{C}$, are 0.20 m apart. Find the magnitude of the force each exerts on the other and state whether it is attractive or repulsive.

2. Three charges lie on the x -axis: $+2.0\ \mu\text{C}$ at $x = 0$, $-2.0\ \mu\text{C}$ at $x = 0.10\ \text{m}$, and $+4.0\ \mu\text{C}$ at $x = 0.30\ \text{m}$. Find the net force (magnitude and direction) on the $+4.0\ \mu\text{C}$ charge.
3. Two identical small spheres carry charges q and $3q$ and repel with force F . They are touched together and separated to the same distance. Find the new force in terms of F .